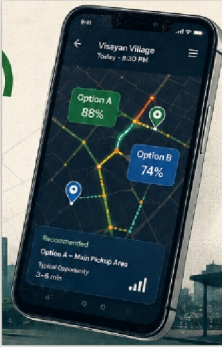


ROUTE SENSE

Know *where* rides are. Know *when* they're available.



- SMARTER DECISIONS**
Find the best places and times to ride.
- REAL-TIME INSIGHTS**
Up-to-date community mobility intelligence.
- CONNECTED JOURNEYS**
Know your ride. Know your mobile signal.
- PRIVACY FIRST**
Aggregated & anonymous on-device data.

1 THE PROBLEM

Knowing a route does not mean knowing if a ride is available right now.

Informal transit changes by time, traffic, and weather. Commuters face daily uncertainty:

- “Can I actually get a ride here right now?”
- “Where is the better place to wait?”
- “How reliable is transportation at this hour?”
- “Will I have mobile signal along the trip?”

2 BENEFICIARIES

PRIMARY USERS

- Students & daily commuters
- Workers & late-night commuters
- Delivery & gig-economy workers

SECONDARY BENEFICIARIES

- Schools, universities & transport coops
- Local government & transport planners
- Researchers & telco mobility planners

3 THE SOLUTION

RouteSense uses built-in passenger smartphone sensors to understand informal transit behavior over time without driver apps or trackers.

- StopSense**
Learns pickup areas, movement, & service probability.
- SignalTrail**
Measures cellular connectivity along transit corridors.
- Where rides are available
- How reliable estimate is
- When they arrive
- Route cellular strength

4 HOW IT WORKS — PASSENGER SENSING TO MOBILITY INTELLIGENCE



5 KEY SYSTEM FEATURES

9 Core Capabilities

88% Service Probability
Live likelihood of finding a ride

Smart Pickup Areas
High-frequency boarding clusters

Opportunity Window
Realistic 3-6 min wait interval

Time-Based Mobility
Morning, night & weekend trends

Confidence & Freshness
31+ observations • High

Connectivity-Aware
Compare speed vs signal strength

Pattern Discovery
Uncovers informal transit routes

Offline Caching
Full query access in dead zones

Zero Driver App
100% passenger crowdsensed

6 EMERGING TECHNOLOGIES & SENSORS

Zero vehicle installation • No driver app or GPS tracker required

- Sensor Fusion & Activity Recognition:** Combines IMU + GNSS to detect *Walking* → *Waiting* → *In Vehicle*.
- Cellular Signal Intelligence:** Maps network QoS during transit journeys.
- Geospatial Clustering & ML:** PostGIS + H3 hexagons with XGBoost predictive analytics.
- Privacy-Preserving On-Device AI:** Local edge inference without raw trip logging.

PHONE SENSORS UTILIZED

- GPS/GNSS
- Accelerometer
- Gyroscope
- Cellular Modem
- Magnetometer

8 REAL-WORLD RECOMMENDATION EXAMPLE

Visayan Village – 8:30 PM

OPTION A: Main Terminal
Service Probability: **88%**
Typical Wait: **3-6 min**
Signal: **Moderate**
FASTEST CHOICE: OPTION A

OPTION B: Alternate Stop
Service Probability: **74%**
Typical Wait: **6-10 min**
Signal: **Very Good**
BEST CONNECTED: OPTION B

7 TECH STACK

- Mobile Client: React Native + TS
- Sensor Layer: Kotlin Core
- Offline Cache: SQLite DB
- Mapping: MapLibre + OSM
- Backend DB: Supabase Postgres
- Spatial Index: PostGIS + Uber H3
- Data Science: Python + Pandas
- Inference: XGBoost ML

9 PROJECT TEAM & PROPONENTS

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- Kyndel Roy Suarez
- Anjelo Vidal

CCE 106 — Application Development & Emerging Technologies | Student Researchers

10 REFERENCES & CITATIONS

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